



Influence of time and temperature on pyrolysis of plastic wastes in a semi-batch reactor

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ABSTRACT

The objective of this work is the study of the influence of temperature and time in the products obtained in the pyrolysis of plastic wastes. The thermal behaviour of a mixture, which resembles municipal plastic wastes has been studied both in a thermogravimetric analyser and in a 3.5 dm³ semi-batch reactor at atmospheric pressure in order to establish the most appropriate time–temperature combination for plastic waste pyrolysis. It has been proved by the authors that temperature has a strong effect in the characteristics of pyrolysis liquids and to a lesser extent in gas and solid properties. At the lowest temperature tested (460 °C), a great proportion of extremely viscous liquids with high content of long hydrocarbon chains are obtained, while at the highest temperature tested (600 °C) low proportion of liquids with a high content of aromatics is produced. The effect of time is not as strong as that of temperature except for very short reaction times (0–15 min). 15–30 min was established as the optimum reaction time range, since total conversion is achieved and longer reaction times do not produce any effect neither in conversion nor in products characteristics. 500 °C has been found to be the most appropriate temperature for the pyrolysis of such plastic waste in terms of both conversion and quality of the products.

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1. Introduction

The production of plastic goods has drastically increased in the last few decades. At present, 60 million tons of such products are produced in Occidental Europe and about 40% of them are estimated to be consumed in packing and packaging services, a short life application which leads to the generation of almost 15 million tons of plastic wastes per year [1]. Nowadays, in Europe, only about 24% of the annually generated plastic wastes coming from household packaging applications are recycled, most of them by means of mechanical processes, while the chemical recycling rate of these wastes is less than 1% and restricted to blast or clinker furnace applications. New processes are needed to increase the chemical recycling percentage, and pyrolysis may be an attractive alternative since it provides an opportunity to obtain valuable liquid and gas fuels from plastic wastes.

In the pyrolysis process (heating in an oxygen free atmosphere), the organic components of the material are decomposed generating liquid and gaseous products, which can be useful as fuels and/or sources of chemicals. The inorganic materials (fillers, metals) remain practically unaltered and free of the binding organic matter; therefore, metals could be separated and the remaining

solid may be reused (additive, filler, pigment) or as a last resort, it would be a minimum waste to be landfilled. Pyrolysis is an especially appropriate recycling technique for waste streams containing different plastics and other materials, for which mechanical recycling is not feasible [2,3].

The pyrolysis of virgin and waste plastics has been intensively studied in the last years [4–8]. Several reactor geometries and experimental configurations have been proved, from micro-pyrolysis and thermogravimetric analysers (TGA) used for analytical pyrolysis studies [4–6] to medium and large scale plants, mainly fluidised bed units, focused on industrial implementation [7,8]. Despite the fact that batch and semi-batch reactors can suffer from temperature gradients due to the low thermal conductivity and high viscosity of plastic wastes, they have also been used to a great extent in lab-scale applications, since they are usually easier to design and operate. Besides, they enable to work with large samples and with great particle sizes, which are closer conditions to those of potential industrial applications.

The influence of operating parameters cannot be easily extrapolated from one installation to another, since such influence usually depends on the specific characteristics of the process. Up to now, temperature has been one of the most studied operating variables, since it is the parameter which most affects thermal cracking of plastics, and as a consequence it has a strong effect on pyrolysis products and on secondary reactions [9–11]. Such reactions are very much influenced by residence time, however, there are few reported studies in the literature about the influence of time on

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pyrolysis products and the few reported studies have been carried out mainly in fluidised bed units [12,13] or in batch reactors (closed autoclaves) [14,15]. In this paper, an experimental study of the influence of time and temperature on pyrolysis of a complex plastic mixture has been carried out in order to establish the optimum time–temperature combination that enables to obtain the best quality products.

On the other hand, most of the above mentioned studies have been carried out with individual plastics or with simple mixtures of very few plastics, which do not resemble complex real plastic wastes. There are almost no pyrolysis studies, which include PVC and PET in the plastic samples, since these two plastics may cause several operating problems such as corrosion or pipeline obstructions. In this paper, the results obtained in pyrolysis of a complex plastic mixture, which includes both PVC and PET in similar proportions to real plastic wastes is presented. A more thorough characterization of pyrolysis products than that usually found in the literature is included, which is essential information to establish the potential applications of pyrolysis products.

2. Materials and methods

2.1. Materials

The plastic mixture which was used for the experiments was composed of the following materials: (1) virgin polyethylene (PE) (Ref. PE-017/PE-071) provided by Repsol Química S.A. and used for household applications, (2) virgin polypropylene (PP) (Ref. PP-040) provided by Repsol Química S.A. and used for general applications, (3) virgin polystyrene (PS) (Ref. HIPS-DL471) provided by Dow Chemical, (4) waste poly(ethylene terephthalate) (PET), washed and milled, coming from recycled bottles and provided by Remaplast S.A., a Spanish company devoted to municipal plastics recycling, and (5) waste poly(vinyl chloride) (PVC) coming from a variety of bottles and provided by Gaiker, a Spanish Technology Centre dedicated to research and innovation in recycling and recovery of plastics among other research areas.

The proportions of the plastics in the mixture were 40 wt% PE, 35 wt% PP, 18 wt% PS, 4 wt% PET and 3% PVC. This composition was established characterizing real samples rejected from an industrial plant located in Amorebieta, in the north of Spain [16,17]. All the plastic materials were used in pellet size (≈ 3 mm) for the pyrolysis experiments; additionally finely ground samples (≤ 1 mm) were prepared for characterization purposes and some TGA analyses.

The characterization results of the plastic mixture and the individual plastic components are presented in Table 1. As it can be seen, the plastic mixture is mainly composed of carbon and hydrogen (84.7 and 12.5 wt% respectively) as a consequence of the high PE, PP and PS content of the sample (93 wt%), materials which are composed just of carbon and hydrogen; this fact also explains its high higher heating value (HHV). It must be mentioned that the sample also contains 1.1 wt% of chlorine, due to the presence of a 32.4 wt% chlorine containing PVC in the mixture.

3. Experimental

The thermal behaviour of the individual plastics was studied using a Mettler Toledo TGA/SDTA851 analyser. The analyses were conducted with 7.5 mg samples, which were heated under nitrogen flow (50 mL/min) to 600 °C at a rate of 20 °C min⁻¹. The temperatures of the maximum degradation rates were determined from the derivative thermogravimetric (DTG) plot. Additionally, thermogravimetric analyses of the mixed sample were carried out with a LECO TGA-500 analyzer. In this case, a total sample mass of 0.5 g was heated at a rate of 20 °C min⁻¹ to the desired temperature and

maintained there for 30 min; nitrogen was passed through at a rate of 4.5 dm³ min⁻¹ during the analysis.

The pyrolysis experiments were carried out using an unstirred stainless steel 3.5 dm³ reactor in semi-batch operation at atmospheric pressure. In a typical run, 100 g of the sample were placed into the reactor and nitrogen was passed through at a rate of 1 dm³ min⁻¹; the system was heated at a rate of 20 °C min⁻¹. In the experiments devoted to the study of temperature, the sample was heated to 460, 500 or 600 °C and maintained in each case at that temperature for 30 min. This time was chosen as standard for the study of the influence of temperature based on previous studies carried out by the authors with other polymeric wastes [18–20]. The influence of time in pyrolysis was explored using 0, 15, 30 and 120 min reaction times and the best temperature selected in the previous series of experiments. Reaction times below 30 min were studied in order to determine if with shorter times total decomposition of the sample could also be achieved as with 30 min, but obviously with the corresponding energy saving. On the other hand, 120 min was chosen as a large enough time to investigate if (1) further reactions take place over 30 min, (2) if solid yields can be reduced to zero and (3) if the products characteristics are modified with longer reaction times. Intermediate times between 30 and 120 min were not explored since the same conversion was obtained at 30 and at 120 min, which indicates that at 30 min the whole sample was already decomposed. No sampling was carried out during these experiments, so the results presented concerning the influence of time in pyrolysis were obtained by analyzing the pyrolysis products collected as a whole after the experiments.

During each run the vapours leaving the reactor flowed to a series of running water cooled gas–liquid separators where the condensed liquids were collected. The uncondensed products were passed through an activated carbon column and collected as a whole in Tedlar plastic bags, to be afterwards tested by gas chromatography. The experimental set-up is presented in Fig. 1. The amount of solids (products in the reactor after pyrolysis) and liquids obtained were weighed, and the pyrolysis yields were calculated as weight percentage with respect to the amount of raw material pyrolysed. Gas yields were as a general rule calculated by difference. Some experiments were specifically devoted to directly quantify the amount of gases by gas chromatography; in such experiments a closure of the mass balance of about 90 wt% was obtained.

The results of the pyrolysis yields which are presented in Section 4 of this paper are the mean value of at least three different pyrolysis runs carried out in the same conditions and which did not differ more than three points in the percentage. After each pyrolysis run, the whole solids, liquids and gaseous products were collected and characterized.

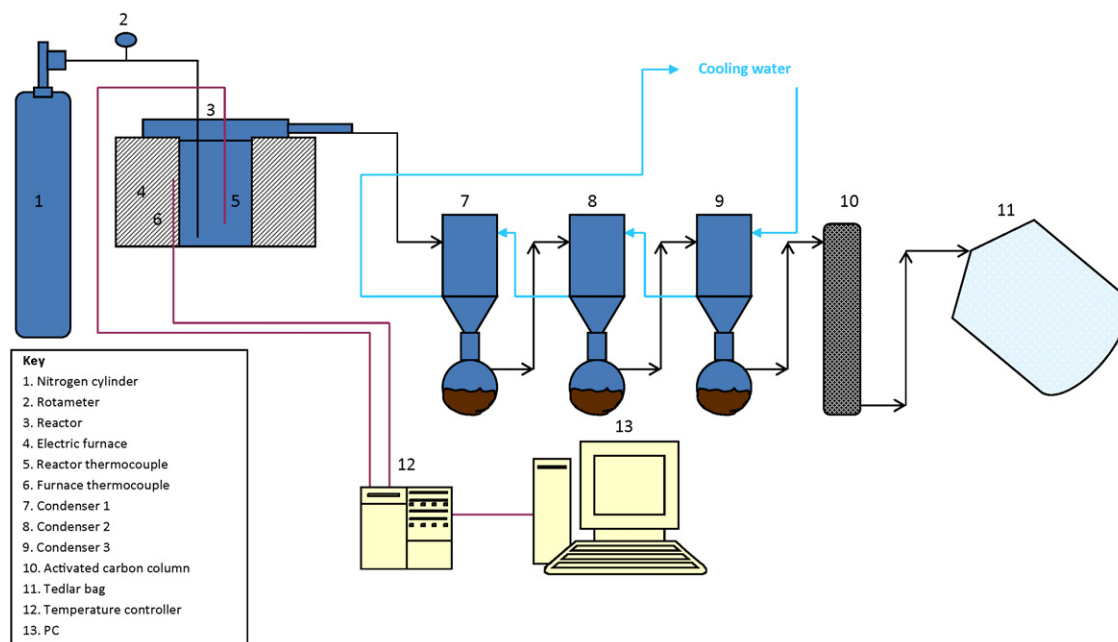
3.1. Analytical techniques

Both the raw materials and the solid and liquid pyrolysis products obtained were thoroughly characterized using the following analytical techniques. The moisture and ash contents of the samples were determined by thermogravimetric analysis according to D3173-85 and D3174-82 ASTM standards respectively, and the elemental composition with an automatic CHN analyser which complies with the ASTM D5373 standard for elemental analysis of fuels. Chlorine was determined following the method 5050 of the Environmental Protection Agency (EPA) of the United States. The higher heating value was determined with an automatic calorimetric bomb complying with the ASTM D3286 standard.

Additionally, pyrolysis liquids were also analysed by gas chromatography coupled with mass spectrometry detector (GC–MS), Agilent 6890 and Agilent 5973 respectively. Characteristics of the method used are shown in Table 2. Identification of the compounds was based on comparison of the retention times with those of cal-

Table 1Moisture, ash and elemental composition (wt%) and HHV (MJ kg⁻¹) of the plastic mixture pyrolysed and the individual components used in the mixture.

Moisture	Ash	C	H	N	Cl	Others ^a	H/C ratio	HHV
0.1	0.0	84.7	12.5	<0.1	1.1	1.5	1.8	43.9

^a By difference.**Fig. 1.** Flow sheet of the experimental set-up used.**Table 2**

Chromatography methods characteristics.

Analysis	GC–MS	GC–TCD/FID
Columns	HP5MS	Molecular sieve 13X Chromosorb 102
Carrier gas	He	He/Ar
Carrier gas flow	1.0 mL/min	48 mL/min
Initial temperature/initial time	40 °C/10 min	40 °C/10 min
Heating rate	8 °C min ⁻¹	6 °C min ⁻¹
Final temperature/final time	280 °C/10 min	200 °C/10 min
Injection temperature	280 °C	110 °C
Injection volume	1.0 µL (split 100:1)	Variable
MS detector temperature (Quad/Source)	150 °C/230 °C	–
TCD detector temperature	–	110 °C
FID detector temperature	–	200 °C

ibration samples and on computer matching against commercial library of mass spectra (Wiley7n) and mass spectra literature data. When the match quality of the identification result provided by the MS search engine was lower than 85%, the result was not considered valid and these compounds are classified as “No identified” in the corresponding tables in this paper. The compounds names correspond to the tentative assignments provided by the MS search engine and have been contrasted, as far as possible, with bibliographic data and occasionally with calibration standards.

Pyrolysis gases were analysed by means of a gas chromatograph (KONIK KNK-3000) coupled with thermal conductivity and flame ionization detectors (GC–TCD/FID). Table 2 shows the characteristics of the method used. Due to the difficulty in distinguishing among isomers from C3 to C6, such discrimination has not been made. The HHV of the gases was calculated according to their composition and to the HHV of the individual components.

4. Results and discussion

4.1. Influence of temperature

4.1.1. Thermogravimetric analyses

Fig. 2 shows the derivative thermogravimetric plot of the individual plastics that constitute the sample. It shows the temperatures at which the maximum degradation rates (peak maximum) take place and also the temperatures at which the decomposition starts (when the peak starts) and finishes (when the peak ends). As expected, PE, PP, PS and PET showed only one peak, which means that they suffer one-step degradation, while in the PVC plot two temperatures of maximum degradation rate can be observed (two peaks), as a consequence of its two-step decomposition mechanism (HCl release corresponds to the first peak and decomposition of the remaining polyene corresponding to the second peak). Additionally, thermogravimetric analyses were also carried out with the mixed sample. Complete decomposition of the sample (100 wt%) was only achieved at 500 °C although from 425 °C pyrolysis can be considered almost complete (99.4–100 wt%). On the contrary, pyrolysis ratio at 400 °C was just 88.2 wt%, which means that a significant amount of sample remained unpyrolysed. For the lab scale pyrolysis plant, 460 °C was selected as the minimum temperature to be tested in order to guarantee complete conversion, taking into account the poor heat transmission of semi-batch reactors.

4.1.2. Pyrolysis yields

The liquid, gas and solid yields (wt%), obtained at the different temperatures are presented in Table 3. It can be seen that while the solid yield has almost a constant value in all the experiments, gas and liquid yields are strongly influenced by temperature, and the higher the temperature, the higher the gas yield and the lower the

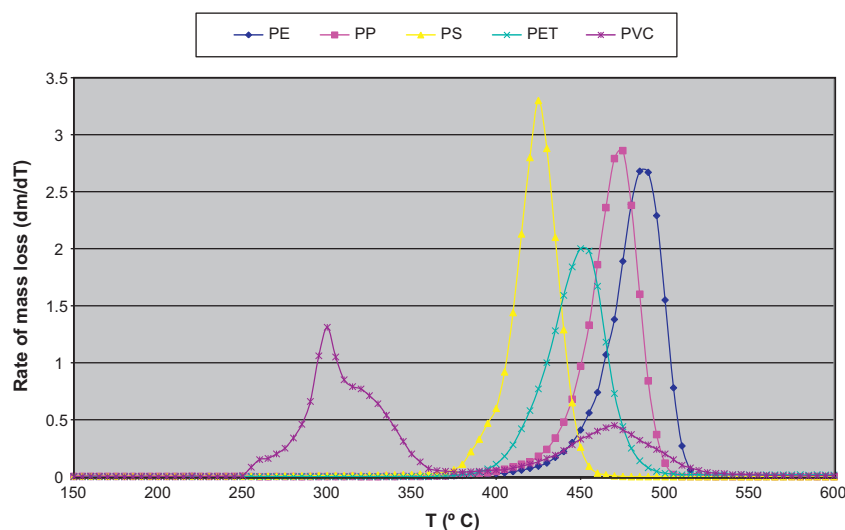


Fig. 2. Derivative thermogravimetric (DTG) plot of the individual plastics used in the mixture.

Table 3

Effect of temperature on pyrolysis yields (wt%).

Temperature (°C)	Liquids	Gases	Solids
460	72.0	26.9	1.1
500	65.2	34.0	0.8
600	42.9	56.2	0.9

liquid yield obtained. This has also been reported by many other authors (e.g. [9,21,22]), and it is attributed to the stronger cracking of C–C bonds that is produced at higher temperatures, which gives rise to lighter hydrocarbons with shorter carbon chains.

Table 3 shows that although the original sample did not contain inorganic matter (see Table 1), in all the experiments a small quantity (0.8–1.1 wt%) of solid products was obtained; this is attributed to char formation, due to secondary repolymerization reactions among the polymer derived products. Char formation in pyrolysis of polymeric wastes is a well documented fact which has been reported and studied before by many research groups (e.g. [23–26]).

4.1.3. Pyrolysis liquids

A summary of the results obtained in GC–MS analysis is presented in Tables 4 and 5. In Table 4 total aromatics (mono and poly aromatics), non-aromatics and unidentified compounds have been quantified, and pyrolysis liquids compounds have been grouped according to their number of carbons (C5–C9, C10–C13 and >C13). In Table 5 the main components of the pyrolysis liquids have been included. It must be mentioned that the values presented in Tables 4 and 5 are % area and not wt%, thus the relative amounts of the different products are not straight reflected by these values due to the response differences of different kinds of organic compounds in MS.

First of all, Table 4 shows that, oddly, in spite of having pyrolysed a plastic mixture mainly composed of long saturated hydrocarbons (PE and PP), no paraffins are obtained in the liquid fraction. This may be explained as follows; when the long polymer chains are cracked, they generate lots of free radical fragments, which need to be stabilised. These radical fragments can further (1) form a double bond generating olefins, (2) combine each other yielding cyclized structures (naphthenes), (3) release hydrogen being later transformed into highly unsaturated products or into aromatics.

The pyrolysis liquids obtained at 460 and 500 °C are mainly composed of aromatics (71.7% and 73.9% area respectively) and unsaturated hydrocarbons (22.3% area in both cases). There are

many references in the literature, which report that aromatics are produced in the pyrolysis of pure polyolefins (e.g. [26–28]), but there are also references, which indicate that aliphatics and naphthenes are the predominant products [9,11]. The fact of the matter is that the formation of aromatics strongly depends on the reactor design and the operating conditions used; when quite high temperatures and long reaction times are used, as in this study, high contents of aromatics are obtained. On the other hand, when complex plastic mixtures are pyrolysed, there are interactions among the polymers derived products, which give rise to the formation of aromatics. Williams and Williams [29] as well as Pinto et al. [30] obtained higher proportions of aromatics in pyrolysis of plastic mixtures than those expected based on the aromatics contents obtained in pyrolysis of the pure components.

It can also be seen in Table 4 that the liquids obtained at 600 °C are almost totally composed of aromatic compounds while there are almost no olefins and naphthenes. The increase in aromatics with temperature has been reported before by other authors (e.g. [7,9,11]), which states that aromatics are formed by means of secondary reactions, which are produced to a major extent at high temperatures. The results in Table 4 indicate that olefin structures are precursors for aromatics formation since their proportion decreases from 22.3% area to 0.4% area as temperature is raised from 460 to 600 °C. However, there is no agreement in the literature about the aromatics formation mechanisms; two main routes have been suggested: Diels–Alder reactions followed by dehydrogenation [31], and unimolecular cyclization reactions followed by dehydrogenation [11], which some authors call “pyrosynthesis”. When complex mixtures are pyrolysed, the combination of both mechanisms is the most probable route to form cyclic structures.

It is also worth noting that the increase in aromatics obtained at 600 °C corresponds mainly to the formation of poly aromatic hydrocarbons (PAH) and their derivatives, since the mono-aromatics contents are similar at all the temperatures. Therefore, it may be stated that PAH are formed by means of secondary reactions which are consequently favoured at high temperatures. It has been proposed that the direct combination of aromatic rings is the mechanism for the formation of PAH at high temperatures [32].

The proportions of the C5–C9, C10–C13 and >C13 fractions of the pyrolysis liquids are also reported in Table 4. C5–C9 is the main fraction in all the pyrolysis experiments; this is a rather convenient result from the point of view of potential applications of these liquids, since C5–C9 is the gasoline carbon number range. This light and highly aromatic fraction should be used blended

Table 4
Aromatic and non-aromatic compounds found in the pyrolysis liquids (% area).

	EXPERIMENT	460 °C	500 °C	600 °C
Aromatics	Mono-aromatics	68.0	69.7	70.8
	Indane derivatives	n.d. ^a	1.1	6.1
	Poly aromatics (PAH) and derivatives	3.7	3.1	22.4
	Total	71.7	73.9	99.3
Non-aromatics	Naphthenes	0.9	n.d. ^a	n.d. ^a
	Lineal olefins	14.4	15.1	0.4
	Branched olefins	7.0	7.2	n.d. ^a
	Total	22.3	22.3	0.4
Unidentified		6.0	3.8	0.3
C5–C9	Aromatics	68.0	69.2	70.6
	Non-aromatics	10.1	10.4	n.d. ^a
	Total	78.1	79.6	70.6
C10–C13	Aromatics	1.2	3.1	23.3
	Non-aromatics	6.3	6.3	0.4
	Total	7.4	9.4	23.8
>C13	Aromatics	2.5	1.7	5.3
	Non-aromatics	6.0	5.6	n.d. ^a
	Total	8.5	7.3	5.3

^a Not detected.

with other non-aromatic petrochemical fractions in order to adjust the aromatic content of the final desired product. >C13 fraction, as expected, decreases as the temperature of the experiment is increased, since the formation of small molecules is favoured at high temperatures. Such effect has also been reported before by other authors [33,34]. It can also be seen that in the liquids obtained in the 600 °C experiment C10–C13 fraction reaches the highest value (23.8% area); the detailed GC–MS results indicate that this value mainly corresponds to PAH and their derivatives which, as it has been shown in Table 4, are quite abundant in the 600 °C liquids.

The main individual components of the pyrolysis liquids are presented in Table 5. For the sake of reduction only those compounds with a percentage quantified area greater than 3% have been included. Additionally, for the purpose of reducing the number of tables, the results of the experiments carried out to study the influence of time have been also included in the tables presented hereinafter (Tables 5–8), but this parameter will be discussed in section 3.2 (in the actual section only the results at 30 min are discussed).

Table 5 shows that in every case, styrene is the most abundant product with percentage areas ranging from 32% to almost 50% area. The next abundant products are toluene (8–17.5% area) and ethyl-benzene (5–8% area). It can be stated that there is some kind of relation among the yields of these three chemicals since the addition of the three yields is quite similar at the three temperatures (62.5%, 61.5% and 58.0% area respectively). Broadly speaking, the tendency is that fluctuations in styrene yield are approximately counterbalanced by toluene and ethyl-benzene yields. Onwudili et al. [15] suggested that toluene and ethyl-benzene may be formed

by the reaction of styrene itself, rather than from the direct degradation of the original sample, which is in agreement with the results obtained in this study. It is also worth noting that styrene yield significantly decreases from 500 to 600 °C which indicates that styrene was formed at lower temperatures and then was decomposed to other chemicals, mainly toluene and ethyl-benzene. The results obtained by other authors in the pyrolysis of polystyrene [15,35] also indicated that the decrease of styrene yield is due to secondary reactions. It is also worth noting the high proportions of naphthalene (6.5% area) and methyl-naphthalene (5.1% area) in the experiment carried out at 600 °C, which contribute to the high proportions of PAH present in the liquid obtained at this temperature. Other PAH found in the liquid fraction in noticeable proportions at this temperature were phenyl-naphthalene (2.9% area), phenyl-benzene (2.5% area) and phenanthrene (1.4% area).

The elemental composition and higher heating value of the pyrolysis liquids obtained in each experiment are presented in Table 6. The carbon and hydrogen content of the liquids follow the same tendency as that observed in the GC–MS analysis; the H/C ratio decreases as the temperature is raised, due to the greater aromatization that is produced at 600 °C. The chlorine content also increases with temperature, which may be explained as follows: at higher temperatures radical fragments are more quickly generated and may have more opportunities to interact with the HCl which evolves from PVC at low temperatures before it leaves the reactor, yielding as a consequence more chlorinated liquids.

Concerning the “others” percentage in the liquids, it most probably corresponds mainly to oxygen derived from PET, since it is an oxygenated polymer. The detailed GC–MS analysis showed

Table 5
Main components of the pyrolysis liquids determined by GC–MS (% area).

Temperature (°C)	460	500				600
Time (min)	30	0	15	30	120	30
Toluene	9.9	11.3	10.7	8.1	8.8	17.5
Dimethyl-heptene	6.7	4.1	4.1	5.9	5.7	n.d. ^a
Ethyl-benzene	7.1	7.6	7.5	5.0	5.7	8.1
Xylenes	<3.0	<3.0	<3.0	<3.0	<3.0	4.5
Styrene	45.5	42.2	42.4	48.4	47.4	32.4
α-Methyl-styrene	3.6	3.8	3.8	4.2	4.7	4.4
Napthalene	<3.0	<3.0	<3.0	<3.0	<3.0	6.5
Methyl-napthalene	n.d. ^a	<3.0	n.d. ^a	n.d. ^a	<3.0	5.1

^a Not detected.

Table 6
Elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis liquids.

Temperature (°C)	Time (min)	C	H	Cl	Others ^a	H/C ratio	HHV
460	30	87.7	11.7	0.4	0.2	1.6	43.5
	0	85.6	11.2	1.0	2.2	1.6	42.5
500	15	87.0	11.1	0.6	1.3	1.5	41.1
	30	86.5	11.3	0.5	1.5	1.6	43.3
	120	86.7	11.3	0.7	1.3	1.6	42.5
600	30	89.2	9.0	1.1	0.7	1.2	40.8

^a By difference.**Table 7**
GC-TCD/FID analysis (wt%) and HHV (MJ kg⁻¹) of pyrolysis gases.

Temperature (°C)	460	500				600
Time (min)	30	0	15	30	120	30
H ₂	0.4	0.2	0.6	0.4	0.5	0.7
CO	1.6	1.6	0.9	0.7	0.6	0.7
CO ₂	2.0	5.4	4.6	2.9	2.7	2.0
Methane	7.9	12.6	11.6	8.3	8.0	13.0
Ethane	10.1	12.8	12.0	10.0	10.1	10.3
Ethene	11.2	12.3	10.5	12.2	12.4	19.3
C3	29.8	28.2	26.0	29.1	29.2	28.2
C4	18.1	14.9	18.8	17.6	17.9	16.3
C5	9.3	9.0	9.5	9.5	9.3	5.3
C6	9.5	2.9	5.5	9.2	9.3	4.2
HHV	48.6	47.2	48.2	48.6	48.9	49.8

Table 8
Moisture, elemental composition (wt%) and HHV (MJ kg⁻¹) of the pyrolysis solids.

Temperature (°C)	Time (min)	Moisture	C	H	Cl	Others ^a	H/C ratio	HHV
460	30	0.1	92.0	3.9	0.1	4.0	0.5	38.5
	0	n.d. ^b	83.8	14.0	0.1	2.2	2.0	47.4
	15	0.4	94.4	3.7	0.2	1.3	0.5	39.4
500	30	0.2	93.7	3.5	0.3	2.3	0.4	38.2
	120	0.3	94.1	3.5	0.1	2.0	0.4	38.2
600	30	0.1	91.7	2.3	0.3	5.6	0.3	36.8

^a By difference.^b Not determined.

that there are some oxygenated compounds in the liquids, such as acetylcyclopentanone and methyl-butanol. However, they have not been included in Table 6 since they are present in very low proportions (<0.5% area). Most of the oxygen derived from PET is transformed into CO and CO₂, as it will be seen in section 3.1.3.

It is worth mentioning that pyrolysis oils have very high HHV, similar to those of conventional liquid fuels, so they may be considered as an appropriate alternative to fossil fuels, since although in terms of energy efficiency the HHV of the oils is comparable to that of the original sample (see Table 1), the advantage of pyrolysis is that it transforms a solid plastic waste into more valuable and easily handled fuels. However, the chlorine contained in the liquids, which is derived from the PVC of the plastic mixture, would probably condition their application as liquid fuels. Different alternatives may be proposed to overcome this problem, such as the use of solid adsorbents mixed with the plastic mixture or the step-wise pyrolysis. In a previous paper published by the authors [36] such alternatives were studied and it was concluded that the chlorine content of the liquids coming from PVC containing plastic mixtures can be drastically reduced by carrying out a low temperature dechlorination step prior to pyrolysis. More exactly, in the mentioned paper it was reported that a reduction of more than 50 wt% of the chlorine in the liquids was achieved and that most of the chlorine was transferred to the gaseous fraction in HCl form.

With a view to select the most appropriate pyrolysis temperature, it must be mentioned that although Tables 4–6 show that

460 °C and 500 °C liquids are quite similar, 460 °C liquids were much more viscous than 500 °C liquids, almost solid at room temperature, and this caused operating problems (pipes obstruction) during the pyrolysis run. On the other hand, the high PAH content of the liquids obtained at 600 °C makes them inappropriate for application from an environmental point of view. Therefore, 500 °C was chosen as the most appropriate temperature for the subsequent study of the influence of time.

4.1.4. Gas composition

Table 7 shows that pyrolysis gases are composed of hydrocarbons ranging from C1 to C6, hydrogen and some carbon dioxide and monoxide. It can be seen that in the 600 °C experiment greater quantities of C1–C3 gases (70.8 wt%) and consequently less C4–C6 gases (25.8 wt%) were produced compared to the experiments carried out at 460 and 500 °C (59.0 and 59.6 wt% respectively of C1–C3 gases and 36.9 and 36.3 wt% respectively of C4–C6 gases). This is quite in accordance with the stronger cracking that is produced at the highest temperature. This tendency to produce lighter hydrocarbons can be clearly observed in methane and ethylene yields, which vary from 7.9 and 11.2 wt% respectively at 460 °C to 13.0 and 19.3 wt% respectively at 600 °C. This behaviour was also observed by Mastral et al. [37] in the pyrolysis of polyethylene in a fluidised bed reactor. On the other hand, it can also be seen that the gases derived from the experiment carried out at 600 °C produced more hydrogen than those derived from the low temperature runs, which can be attributed to the hydrogen release reactions that are pro-

duced in the formation of aromatics, which is favoured at high temperatures.

It is also worth noting that the HHV of all the gases is in the range of that of natural gas (48–53 MJ kg⁻¹). For this reason pyrolysis gases may be used as gaseous fuels to supply the energetic demand of the process, and the surplus may be valorised.

4.1.5. Pyrolysis solids

The composition of the pyrolysis solids is presented in Table 8. The pyrolysis solids hardly differ one another, and are mainly composed of carbon, which corresponds to the carbonaceous product (char) previously mentioned which is formed during pyrolysis, some hydrogen and a slightly variable quantity of other elements. A slight decrease of the H/C ratio with temperature can be observed; it is a consequence of the stronger carbonization that is produced at higher temperatures. The pyrolysis solids are in all cases carbonaceous materials with very high HHV, which can be used as an alternative to fossil solid fuels.

4.2. Influence of time

As it has been mentioned before, 500 °C was considered the most appropriate temperature to carry out the experiments devoted to study the influence of time on plastic wastes pyrolysis. The effect of reaction time was explored in the range 0–120 min; the reaction time was counted from the moment the experiment temperature was reached. The results obtained are shown below.

4.2.1. Pyrolysis yields

The evolution of the liquid, gas and solid yields (wt%) as a function of time is presented in Fig. 3. It can be seen that for 0 min a significant conversion to gases and liquids (75.9 wt%) is produced, which has taken place just during the heating and cooling stages of the process, due to the high thermal inertia of the reaction system. From this time on, solid yield quickly decreases and remains practically constant over 15 min, which indicates that complete decomposition of the sample has been achieved; the solid yield is very low (≈1 wt%) and corresponds to char formed during the process, and it cannot be further decomposed whatever the reaction time is. Concerning liquid yield, it increases from 0 to 30 min, reaching the highest value (65.2 wt%) at 30 min; no further increase in liquid yield was observed over 30 min. Similar effect of time on liquid yields was found by Lee [38] who showed that once total pyrolysis is achieved, liquid yield remains constant regardless of the reaction time used.

4.2.2. Pyrolysis liquids

The results obtained in the GC–MS analysis of the liquids are presented in Figs. 4 and 5. All the identified compounds have been grouped in: aromatics, lineal olefins and branched olefins. Additionally, these compounds have also been grouped according to their number of carbons: C5–C9, C10–C13 and >C13 compounds.

Fig. 4 shows that non-aromatics increase from 0 to 30 min and then remains constant. Consequently, the maximum aromatics yield was obtained in the 0 min experiment (80.1% area). Lee and Shin [39] obtained 60% of aromatics in the initial stages of the reaction in pyrolysis of PE, PP and PS mixtures at 350 °C. These authors suggested that such high aromatics content was due to PS decomposition, which takes place at lower temperatures than polyolefins (see Fig. 2) and produces high quantities of aromatics. The results obtained in this paper cannot only be explained by such theory because at 500 °C and 0 min as much as 75.9 wt% of the initial sample was pyrolysed, and the proportion of PS in the sample was only 18 wt%, so the aromatics come from some other materials than PS and they must be produced by secondary reactions that take place

at the early steps of the process. According to the thermal decomposition temperature of the plastics used in the experiments (Fig. 2), the plastic mixture which was left in the reactor in the 0 min run must be mainly composed of PE and some PP, and this may be reason why olefins increase from 0 to 30 min, since in this period only PE and PP are decomposed and no interactions with the other plastics are produced.

Fig. 5 shows that there is only a slight influence of time in the different carbon number fractions yields. While C5–C9 and C10–C13 fractions remain practically unaltered, a slight increase in the >C13 fraction can be observed. The increase in long chain compounds with time may be attributed to the fact that the plastics or their derived products, which remain more time in the reactor decompose generating heavier products than those generated in the early stages of the process.

Table 5 shows the main chemicals found in the pyrolysis liquids. A similar distribution as that obtained in the previous experiments carried out at different temperatures and 30 min can be seen: styrene, toluene and ethyl-benzene are the main components. It can be seen that at the longer reaction times (30 and 120 min) more styrene, α-methyl-styrene and dimethyl-heptene while less toluene and ethyl-benzene are produced than at the shorter reaction times (0 and 15 min). Therefore, the decomposition reactions that take place in the later stages of the process yield a comparatively higher proportion of styrene, α-methyl-styrene and dimethyl-heptene than the early stages decomposition reactions.

The elemental composition and higher heating value of the pyrolysis liquids obtained at 500 °C for different reaction times are presented in Table 6. It can be seen that there is almost no influence of time in the elemental compositions of the liquids, which are mainly composed of carbon (85–87 wt%) and hydrogen (≈11 wt%) for every reaction time. In this case, the highest chlorine content was found in the 0 min experiment liquids, due to the fact that at the early stages of the process all the chlorine has already released from the sample (Fig. 2) while the final liquid yield has not yet been achieved, being the chlorine content proportionally higher in the 0 min experiment liquids. Pyrolysis oils have for all the reaction times very high HHV.

4.2.3. Gas composition

The composition of the pyrolysis gases is presented in Table 7. It can be observed that the proportion of light hydrocarbons, as methane and ethane, as well as that of the oxygenated compounds CO and CO₂, decrease with time. This fact suggests that these compounds are formed in a greater proportion in the first steps of the pyrolysis process. On the other hand, the heavier gases (C4–C6) yields clearly show an increase from 0 (26.8 wt%) to 120 min (36.4 wt%). The increase in heavy compounds with time was also observed in the pyrolysis liquids (Fig. 5) and it can be also explained by the above mentioned theory concerning the fact that the plastics or their derived products which remain more time in the reactor decompose generating heavier products. The HHV of the 0 min run was a little bit lower than those of the other gases, which may be attributed to its higher CO₂ content. Anyhow, the HHV (47–49 MJ kg⁻¹) was in all cases in the range of that of the natural gas (48–53 MJ kg⁻¹).

4.2.4. Pyrolysis solids

The composition of the pyrolysis solids obtained at the different reaction times is presented in Table 8. It can be seen that the contents of carbon (83.8 wt%) and hydrogen (14.0 wt%) of the solid obtained in the 0 min run are quite similar to those of polyolefins. Therefore, the unconverted product of that run must be mainly composed of PE and PP; this is corroborated by the TGA analysis of the individual plastics (Fig. 2), which showed that the decomposition of these plastics takes place at somewhat higher temperatures

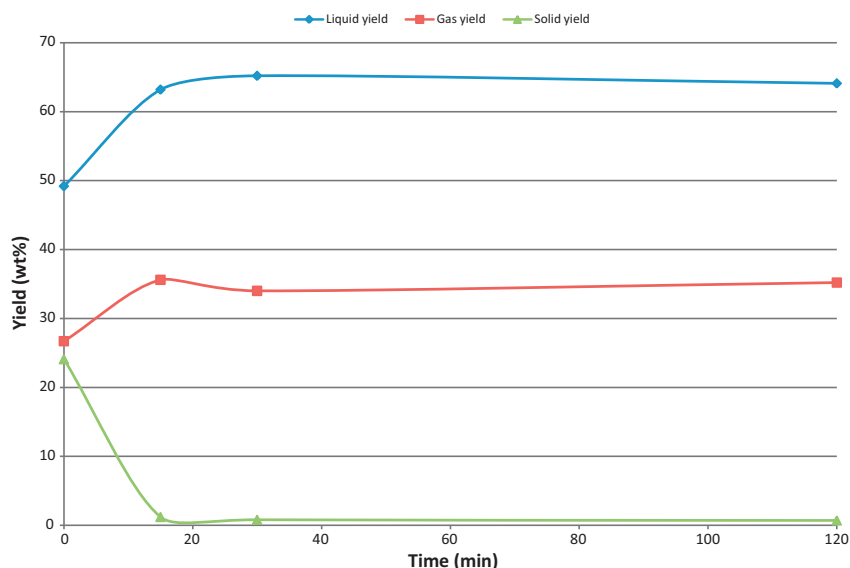


Fig. 3. Pyrolysis yields (wt%) as a function of time at 500 °C.

than those of the other plastics in the mixture. On the contrary, the solids obtained at 15, 30 and 120 min are mainly composed of carbon, which indicates that it is char, the carbonised product that is usually formed in pyrolysis of many plastics; it can also be seen that they hardly differ from one another. Concerning HHV, Table 8 shows that it is higher for the solids of the 0 min run, which is due to the fact that this solid contains a greater proportion of hydrogen, which has a much higher heat of combustion than carbon.

4.3. Energy balance analysis

A theoretical energy balance per unit weight of plastic has been developed for the pyrolysis process at 500 °C and 30 min. The energy fluxes that have been considered are the following; the input energy (required energy) includes the sensible heat of the sample from ambient temperature to 500 °C (Q_s) and the energy needed to break down the polymer bonds (Q_b). As output energy it has only been considered the energy that can be provided by the

gaseous fraction, calculated as a function of its HHV (Q_g). In this way, the general balance equation is:

$$Q_s + Q_b = Q_g \quad (1)$$

4.3.1. Input energy calculation

The sensible heat of the sample in the heating process is:

$$Q_s = C_p \times \Delta T \quad (2)$$

where C_p is the specific heat of the sample and ΔT is the temperature variation from 20 to 500 °C (480 °C). C_p has been calculated from the specific heat of each of the plastics, which compose the sample [40] and taking into account the proportions of each of them in this sample. In this way, Q_s is estimated to be 885 kJ kg⁻¹.

On the other hand, the energy needed to break down the polymer bonds is:

$$Q_b = n \times Mw \times Eb \quad (3)$$

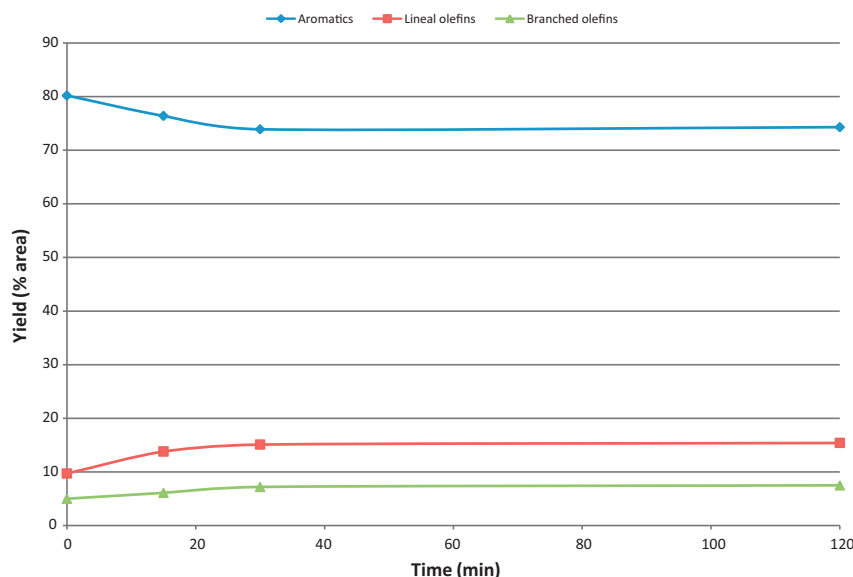


Fig. 4. Aromatics, lineal olefins and branched olefins yields (area %) as a function of time at 500 °C.

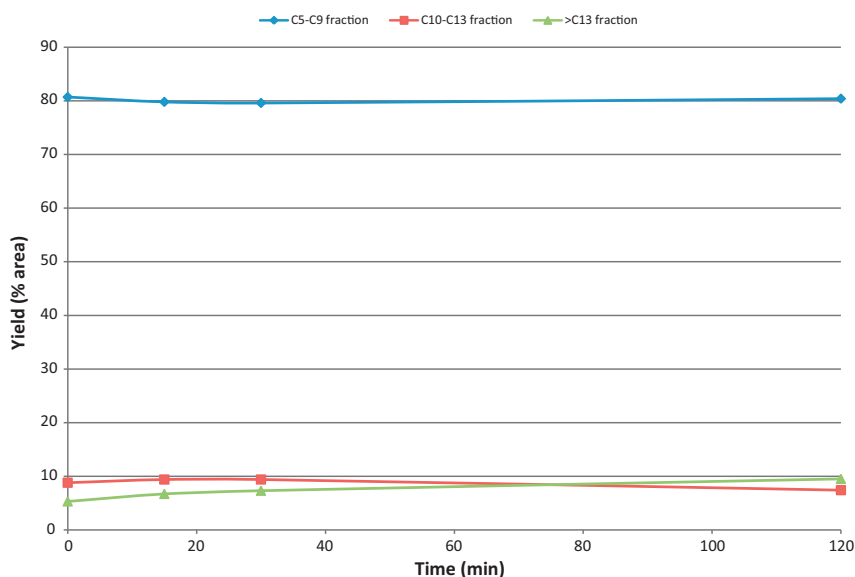


Fig. 5. C5–C9, C10–C13 and >C13 fractions (area %) as a function of time at 500 °C.

where M_w is the average molecular weight of the polymeric sample, E_b the C–C bond dissociation energy, and n the number of bonds which must be broken in the decomposition process. The following assumptions have been considered in the calculation of Q_b : (1) an average molecular weight of $100,000 \text{ g mol}^{-1}$, which is within the order of magnitude of packaging plastics molecular weight [41,42], (2) only C–C bond ruptures have been considered and with a dissociation energy of 350 kJ mol^{-1} and (3) the number of bonds which must be broken has been calculated from the division between the average molecular weight of the sample and the average molecular weight of the pyrolysis products (liquids + gas), which has been established in $\approx 100 \text{ g mol}^{-1}$ taken into account the composition data shown in Tables 5 and 7. In this way, n is ≈ 1000 , Q_b is $\approx 3500 \text{ kJ kg}^{-1}$ and therefore the total input energy ($Q_s + Q_b$) is $\approx 4400 \text{ kJ kg}^{-1}$.

4.3.2. Output energy calculation

In order to calculate the output energy per unit weight of plastic, the HHV of the pyrolysis gases (48.6 MJ kg^{-1}) as well as the pyrolysis gas yield (34.0 wt%), have to be considered. This way, the output energy is $17,000 \text{ kJ kg}^{-1}$. It has been reported that the electrical efficiency of internal combustion engines is in the range 35–40% [43]. In order to be more specific, the obtained in a tyre pyrolysis plant assuming that the gas produced in such plant has the same HHV of that obtained in this work has been calculated, taking into account the energy produced by the combustion engines of this tyre pyrolysis plant [44]; this electrical efficiency is $\approx 22.5\%$. In the present study an average electrical efficiency of 30% has been considered and consequently the Q_g turns out to be 5100 kJ kg^{-1} .

As a result of the input and output energy calculation, there is a positive energy balance ($+700 \text{ kJ kg}^{-1}$) which indicates that pyrolysis of plastic waste could be energetically sustainable by using just the pyrolysis gases as a source to supply the energetic demand of the process and that there may be a surplus which may be used for additional power generation. Besides, it has to bear in mind that the solid and liquid fraction of the process could be also used for energy generation if needed.

5. Conclusion

Pyrolysis is an attractive alternative for recycling mixed plastic waste. Conversions to liquids and gases as high as 99 wt% are

obtained. The liquids may be used as high HHV alternative fuels or as a source of valuable chemicals, such as styrene or toluene. Gases can be used to supply the energetic demand of the process and the surplus may be used for additional power generation, which indicates that plastic waste pyrolysis is an energetically sustainable process.

As pyrolysis temperature is raised, gas yields significantly increase to the detriment of liquid yields. 460°C is the lower temperature at which total conversion is achieved but the liquids are extremely viscous (semi-solid at room temperature) and difficult to handle. 500°C was established as the optimal temperature for plastic waste pyrolysis, in terms of both conversion and quality of the pyrolysis liquids. Reaction time in the range 15–30 min is enough to achieve total conversion of the plastic waste.

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